EXPERIMENT 2

Demonstrate the statistics for a sample data like mean, standard deviation, normal/uniform distribution, variance and correlation.

AIM

To demonstrate basic descriptive statistics including mean, variance, standard deviation, and correlation using the Students Performance dataset, and visualize the distributions.

PROCEDURE

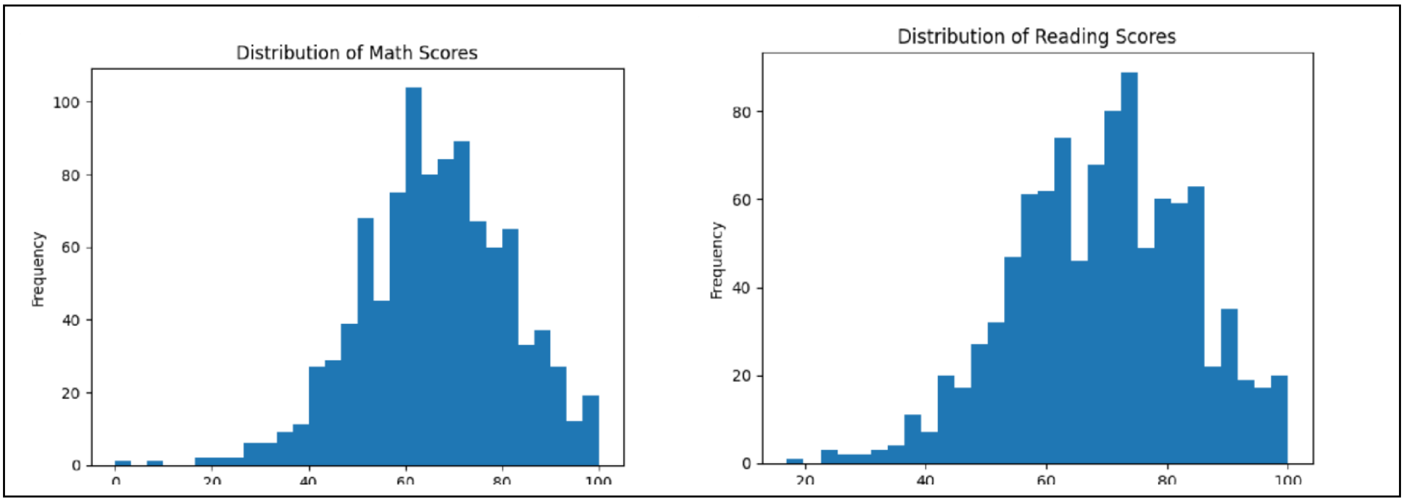
1. Import pandas, matplotlib, and math libraries.
2. Load the StudentsPerformance.csv dataset using pd.read_csv().
3. Extract math scores and reading scores as arrays.
4. Compute the mean of math scores manually using a loop.
5. Compute the variance of math scores manually.
6. Compute the standard deviation using math.sqrt().
7. Plot histogram of math score distribution.
8. Plot histogram of reading score distribution.
9. Compute the mean of reading scores.
10. Calculate the Pearson correlation coefficient between math and reading scores manually.
11. Visualize the correlation using a scatter plot.

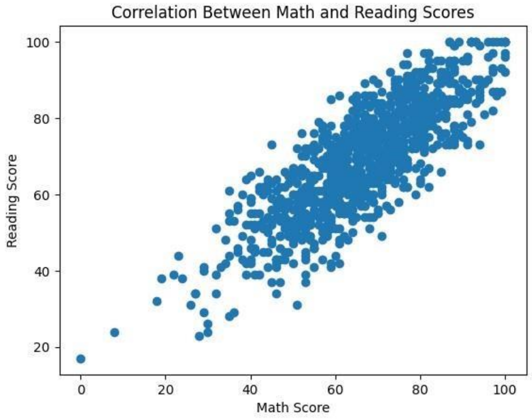
CODE
```
import pandas as pd
import matplotlib.pyplot as
plt import math

df = pd.read_csv('StudentsPerformance.csv') df.head()

math_scores = df['math score'].values
reading_scores = df['reading score'].values n
= len(math_scores)

# Mean sum_math = 0 for x
in math_scores: sum_math
+= x  mean_math =
sum_math / n

# Variance var_sum = 0 for x in
math_scores:     var_sum += (x -
mean_math) ** 2 variance_math = var_sum
/ n

# Standard Deviation  std_math =
math.sqrt(variance_math)

# Histogram of Math Scores plt.hist(math_scores,
bins=30)   plt.title('Distribution    of    Math
```

```
Scores')        plt.xlabel('Math        Score')
plt.ylabel('Frequency') plt.show()
# Mean of Reading Scores
sum_reading = 0 for r in
reading_scores: sum_reading
+= r mean_reading
= sum_reading / n

# Pearson Correlation num = 0; den_x = 0; den_y = 0 for i in range(n): num +=
(math_scores[i] - mean_math) * (reading_scores[i] - mean_reading) den_x +=
(math_scores[i] - mean_math) ** 2       den_y += (reading_scores[i] -
mean_reading) ** 2 correlation = num / math.sqrt(den_x * den_y)

# Scatter Plot plt.scatter(math_scores, reading_scores)
plt.xlabel('Math Score') plt.ylabel('Reading Score')
plt.title('Correlation Between Math and Reading Scores') plt.show()
```

OUTPUT

```
Mean (Math):       66.089
Variance (Math):   229.6897900000004  Std
Dev (Math):  15.155496659628165
Correlation (Math vs Reading): 0.8175796636720533
```





RESULT

The mean math score was 66.089 with a standard deviation of 15.15. A strong positive Pearson correlation of 0.818 was found between math and reading scores, suggesting students who perform well in math also tend to perform well in reading.